Mathematical Statistics

Lecture 6: An Introduction to Bayes' Theorem

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Contents

1 Foundations: Revisiting Conditional Probability

Before diving into Bayes' Theorem, we must first recall the concept of **conditional probability**.

- **Definition:** Conditional probability measures the probability of an event occurring, given that another event has already occurred.
- **Notation:** The probability of event A occurring given that event B has occurred is written as $P(A|B)$.
- **Formula:**
$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$
  Where $P(A \cap B)$ is the probability that both events A and B occur (the intersection of A and B).

1.1 The Multiplication Rule

By rearranging the formula for conditional probability, we get a crucial relationship known as the multiplication rule:

$$P(A \cap B) = P(A|B) \times P(B)$$

This states that the probability of both A and B happening is the probability of B happening multiplied by the probability of A happening, given that B has already happened. This rule is the foundation upon which Bayes' Theorem is built.

2 The Motivation: Forward vs. Inverse Problems

In many real-world scenarios, we can easily measure the probability of an *effect* given a *cause*, but what we truly want to know is the probability of the *cause* given an observed *effect*. This is the core of an **inverse problem**.

Forward vs. Inverse Problems in Medical Diagnosis

Forward Problem (Easier to determine): What is the probability of a patient having a specific symptom (e.g., a positive test result, **Event A**) given that they have a known disease (e.g., cancer, **Event B**)?

$$P(\text{Symptom}|\text{Disease}) \quad \text{or} \quad P(A|B)$$

This can be determined through clinical trials by observing a group of patients with the known disease.

Inverse Problem (More useful for a doctor): What is the probability that a patient has the disease **(Event B)** given that they are presenting a specific symptom **(Event A)**?

$$P(\text{Disease}|\text{Symptom}) \quad \text{or} \quad P(B|A)$$

This is what a doctor needs to make a diagnosis. We reason backward from an observable effect (the symptom) to an unobservable cause (the disease).

Bayes' Theorem provides the mathematical framework to solve these inverse problems.

3 Deriving and Understanding Bayes' Theorem

3.1 The Derivation

The derivation is surprisingly simple and flows directly from the symmetry of the multiplication rule.

1. We know that the probability of A and B occurring is symmetrical, so $P(A \cap B) = P(B \cap A)$.

2. We can express this intersection in two ways using the multiplication rule:

$$P(A \cap B) = P(A|B) \times P(B)$$
$$P(B \cap A) = P(B|A) \times P(A)$$

3. Setting these two expressions equal to each other gives:

$$P(B|A) \times P(A) = P(A|B) \times P(B)$$

4. To find our desired inverse probability, $P(B|A)$, we simply divide by $P(A)$:

$$P(B|A) = \frac{P(A|B) \times P(B)}{P(A)}$$

3.2 The Terminology of Bayesian Inference

In the field of Bayesian statistics, each part of the theorem has a specific name that reflects its role in updating our beliefs.

Anatomy of Bayes' Theorem

$$\underbrace{P(B|A)}_{\text{Posterior}} = \frac{\overbrace{P(A|B)}^{\text{Likelihood}} \times \overbrace{P(B)}^{\text{Prior}}}{\underbrace{P(A)}_{\text{Evidence}}}$$

Posterior Probability $P(B|A)$: The *updated* probability of event B occurring after we have accounted for the new evidence from event A. This is what we calculate.

Prior Probability $P(B)$: Our *initial* belief or probability of event B occurring *before* considering any new evidence from A.

Likelihood $P(A|B)$: The probability of observing our evidence A *given that* our hypothesis B is true. This term "updates" our prior belief.

Marginal Likelihood (Evidence) $P(A)$: The total probability of observing the evidence A, regardless of B. It acts as a normalization constant.

4 Different Forms of Bayes' Theorem

The denominator, $P(A)$, can sometimes be difficult to calculate directly. We can expand it using the **Law of Total Probability**.

Form 1: The Simple Form Used when $P(A)$ is known.

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}$$

Form 2: Expanded Form (with two outcomes) The most common form, where an event either happens (B) or it doesn't (B^c, the complement).

$$P(B|A) = \frac{P(A|B)P(B)}{P(A|B)P(B) + P(A|B^c)P(B^c)}$$

Form 3: The General Form Used when there are multiple, mutually exclusive events $B_1, B_2, \ldots, B_n$.

$$P(B_j|A) = \frac{P(A|B_j)P(B_j)}{\sum_{i=1}^{n} P(A|B_i)P(B_i)}$$

5 Worked Example: Cancer Screening

This example demonstrates the power and counter-intuitive results of Bayes' Theorem.

Medical Diagnosis Problem

Problem Statement: A specific type of cancer is very rare, affecting **0.1%** of the population. A screening test for this cancer is **99% accurate** (i.e., it correctly identifies both the presence and absence of the disease 99% of the time).
Question: If a randomly selected person tests positive, what is the actual probability that they have the cancer?

1. **Define Events:**
   - C: The person has cancer.
   - C^c: The person does not have cancer.
   - Pos: The test result is positive.
2. **Identify Known Probabilities:**
   - $P(C) = 0.001$ (Prior probability of having cancer).
   - $P(C^c) = 1 - 0.001 = 0.999$.
   - $P(Pos|C) = 0.99$ (True Positive Rate: test is positive if you have cancer).
   - $P(Pos|C^c) = 1 - 0.99 = 0.01$ (False Positive Rate: test is positive even if you don't have cancer).
3. **State the Goal:** Find the posterior probability, $P(C|Pos)$.
4. **Apply Bayes' Theorem (Form 2):**

$$P(C|Pos) = \frac{P(Pos|C)P(C)}{P(Pos|C)P(C) + P(Pos|C^c)P(C^c)}$$

5. **Substitute Values and Calculate:**

$$\begin{aligned} P(C|Pos) &= \frac{(0.99) \times (0.001)}{(0.99) \times (0.001) + (0.01) \times (0.999)} \\ &= \frac{0.00099}{0.00099 + 0.00999} \\ &= \frac{0.00099}{0.01098} \\ &\approx 0.09016 \end{aligned}$$

6. **Interpret the Result:** Even with a 99% accurate test, a positive result means there is only about a **9% chance** the person actually has cancer. Over 90

Summary and Key Takeaways

- **Core Idea:** Bayes' Theorem is a formal method for updating a probability (prior) based on new evidence to arrive at a revised probability (posterior).
- **Solves Inverse Problems:** It allows us to calculate $P(\text{Cause}|\text{Effect})$ when we only know $P(\text{Effect}|\text{Cause})$.
- **Real-World Impact:** It is a cornerstone of modern statistics, machine learning (e.g., spam filters), medical diagnostics, and any field involving inference under uncertainty.

6 Practice Exercises

1. **Spam Filtering:** A spam filter is 95% accurate at identifying spam ($P(\text{Flagged}|\text{Spam}) = 0.95$). It also has a false positive rate of 2% ($P(\text{Flagged}|\text{Not Spam}) = 0.02$). If 80
2. **Factory Quality Control:** A factory has two machines, Machine X and Machine Y. Machine X produces 60% of the daily output, and Machine Y produces 40%. 3% of items from Machine X are defective, while 5% of items from Machine Y are defective. If an item selected at random is found to be defective, what is the probability it was produced by Machine X?

7 Solutions to Exercises

7.1 Solution to Exercise 1 (Spam Filtering)

- **Events:** S = Spam, S^c = Not Spam, F = Flagged as Spam.
- **Probabilities:** $P(S) = 0.80$, $P(S^c) = 0.20$, $P(F|S) = 0.95$, $P(F|S^c) = 0.02$.
- **Goal:** Find $P(S^c|F)$.
- **Calculation:**

$$\begin{aligned} P(S^c|F) &= \frac{P(F|S^c)P(S^c)}{P(F|S)P(S) + P(F|S^c)P(S^c)} \\ &= \frac{(0.02) \times (0.20)}{(0.95) \times (0.80) + (0.02) \times (0.20)} \\ &= \frac{0.004}{0.76 + 0.004} = \frac{0.004}{0.764} \approx 0.0052 \end{aligned}$$

- **Answer:** There is approximately a 0.52% chance that an email flagged as spam is actually legitimate.

7.2 Solution to Exercise 2 (Factory Quality Control)

- **Events:** X = From Machine X, Y = From Machine Y, D = Defective.
- **Probabilities:** $P(X) = 0.60$, $P(Y) = 0.40$, $P(D|X) = 0.03$, $P(D|Y) = 0.05$.
- **Goal:** Find $P(X|D)$.
- **Calculation:**

$$\begin{aligned} P(X|D) &= \frac{P(D|X)P(X)}{P(D|X)P(X) + P(D|Y)P(Y)} \\ &= \frac{(0.03) \times (0.60)}{(0.03) \times (0.60) + (0.05) \times (0.40)} \\ &= \frac{0.018}{0.018 + 0.020} = \frac{0.018}{0.038} \approx 0.4737 \end{aligned}$$

- **Answer:** There is approximately a 47.4% chance that the defective item was produced by Machine X.

8 The Core Idea: Solving Inverse Problems

Recall that Bayes' Theorem is our primary tool for solving **inverse problems**. It helps us answer questions like:

> "What is the probability that a person is a drug user, given that they had a positive test result?"

This lecture explores this inverse problem through the practical lens of workplace drug screening, demonstrating how our intuition can often be misleading without a rigorous Bayesian analysis.

9 Scenario 1: A Test with Symmetrical Accuracy

Let's begin with a hypothetical example where the test's accuracy is the same for users and non-users.

Problem: Symmetrical 90% Accuracy

Problem Setup:

- A company screens its employees for a specific drug.
- **10%** of the population being tested are actual users of the drug.
- The test used is **90% accurate**. This means it correctly identifies a user 90% of the time and correctly identifies a non-user 90% of the time.

Question: If an employee's test comes back positive, what is the probability that they are actually a user of the drug?

Step-by-Step Solution

1. **Define the Events:**
   - U: The person is a User of the drug.
   - U^c: The person is a Non-User.
   - Pos: The test result is Positive.
2. **Translate the Problem into Probabilities:**
   - **Prior Probability:** $P(U) = 0.10 \implies P(U^c) = 0.90$.
   - **True Positive Rate:** $P(Pos|U) = 0.90$.
   - **True Negative Rate:** $P(Neg|U^c) = 0.90$.
   - **False Positive Rate:** From the above, we infer $P(Pos|U^c) = 1 - 0.90 = 0.10$.
3. **Apply Bayes' Theorem:** We want to find $P(U|Pos)$.

$$P(U|Pos) = \frac{P(Pos|U)P(U)}{P(Pos|U)P(U) + P(Pos|U^c)P(U^c)}$$

4. **Substitute Values and Calculate:**

$$\begin{aligned} P(U|Pos) &= \frac{(0.90) \times (0.10)}{(0.90) \times (0.10) + (0.10) \times (0.90)} \\ &= \frac{0.09}{0.09 + 0.09} = \frac{0.09}{0.18} = 0.50 \end{aligned}$$

5. **Interpret the Result:** The probability that an employee is a user, given a positive test, is only **50%**. This is no better than a coin flip. The number of false positives from the large non-user pool equals the number of true positives from the small user pool.

10 Scenario 2: Introducing Sensitivity and Specificity

In reality, a test's accuracy often isn't symmetrical. This introduces two crucial concepts for evaluating any diagnostic test.

Key Terminology

Sensitivity (True Positive Rate): The ability of a test to correctly identify those who **have** the condition.

$$\text{Sensitivity} = P(\text{Positive Test}|\text{Is a User}) = P(Pos|U)$$

Specificity (True Negative Rate): The ability of a test to correctly identify those who do **not have** the condition.

$$\text{Specificity} = P(\text{Negative Test}|\text{Is a Non-User}) = P(Neg|U^c)$$

A good test must balance both. A 100% sensitive test can be made by always flagging positive, but it would have 0% specificity and be useless.

Problem: Asymmetrical Accuracy

New Problem Setup: Let's modify the test's accuracy while keeping the user rate at 10%.

- **Sensitivity:** The test is **90%** sensitive. $P(Pos|U) = 0.90$.
- **Specificity:** The test is only **80%** specific. $P(Neg|U^c) = 0.80$.

This means the test now has a higher false positive rate.

Step-by-Step Solution

1. **Update the Probabilities:**
   - $P(U) = 0.10$ and $P(U^c) = 0.90$ (Unchanged).
   - Sensitivity: $P(Pos|U) = 0.90$ (Unchanged).
   - Specificity: $P(Neg|U^c) = 0.80$.
   - **New False Positive Rate:** $P(Pos|U^c) = 1 - 0.80 = 0.20$.
2. **Re-apply Bayes' Theorem:** The formula remains the same.
$$P(U|Pos) = \frac{P(Pos|U)P(U)}{P(Pos|U)P(U) + P(Pos|U^c)P(U^c)}$$
3. **Calculate the New Result:**
$$\begin{aligned} P(U|Pos) &= \frac{(0.90) \times (0.10)}{(0.90) \times (0.10) + (0.20) \times (0.90)} \\ &= \frac{0.09}{0.09 + 0.18} = \frac{0.09}{0.27} = \frac{1}{3} \approx 0.333 \end{aligned}$$
4. **Interpret the New Result:** With a less specific test, the probability that a person with a positive test is an actual user drops to **33.3%**. Now, two out of every three positive tests are wrong.

Summary and Key Takeaways

- **Context is Everything:** A test's accuracy score is misleading without knowing the base rate (prior probability) of the condition.
- **Rare Events Generate False Positives:** When testing for a rare condition, even a small error rate applied to a large "negative" population can generate a large number of false positives, often overwhelming the true positives.
- **Sensitivity vs. Specificity:** Understanding both is crucial for evaluating a test's real-world utility. High specificity is critical for avoiding false positives.
- **Application to Machine Learning:** This same logic applies to training algorithms to detect rare events (e.g., fraud detection). Overall accuracy is often a poor metric; sensitivity and specificity are better.

11 Practice Exercises

1. **Airport Security:** An alarm system is 98% sensitive ($P(\text{Alarm}|\text{Threat}) = 0.98$) and 95% specific ($P(\text{No Alarm}|\text{No Threat}) = 0.95$). If only 1 in 1,000 items (0.1%) poses a genuine threat, what is the probability that an item that triggers the alarm is actually a threat?
2. **Email Spam Filter:** A spam filter is 99% sensitive ($P(\text{Flagged}|\text{Spam}) = 0.99$) and 99.5% specific ($P(\text{Not Flagged}|\text{Not Spam}) = 0.995$). If 15% of your emails are spam, what is the probability that an email placed in your inbox (not flagged) is actually spam?

12 Solutions to Exercises

12.1 Solution to Exercise 1 (Airport Security)

- **Events:** T = Threat, T^c = No Threat, A = Alarm.
- **Probabilities:** $P(T) = 0.001$, $P(T^c) = 0.999$, $P(A|T) = 0.98$, $P(A|T^c) = 1 - 0.95 = 0.05$.
- **Goal:** Find $P(T|A)$.
- **Calculation:**

$$\begin{aligned} P(T|A) &= \frac{P(A|T)P(T)}{P(A|T)P(T) + P(A|T^c)P(T^c)} \\ &= \frac{(0.98)(0.001)}{(0.98)(0.001) + (0.05)(0.999)} \\ &= \frac{0.00098}{0.00098 + 0.04995} = \frac{0.00098}{0.05093} \approx 0.0192 \end{aligned}$$

- **Answer:** There is only a **1.92%** chance that an item triggering the alarm is a real threat.

12.2 Solution to Exercise 2 (Email Spam Filter)

- **Events:** S = Spam, S^c = Not Spam, F = Flagged, F^c = Not Flagged.
- **Probabilities:** $P(S) = 0.15$, $P(S^c) = 0.85$, $P(F|S) = 0.99$, $P(F^c|S^c) = 0.995$.
- We also need the False Negative Rate: $P(F^c|S) = 1 - P(F|S) = 1 - 0.99 = 0.01$.
- **Goal:** Find $P(S|F^c)$.
- **Calculation:**

$$\begin{aligned} P(S|F^c) &= \frac{P(F^c|S)P(S)}{P(F^c|S)P(S) + P(F^c|S^c)P(S^c)} \\ &= \frac{(0.01)(0.15)}{(0.01)(0.15) + (0.995)(0.85)} \\ &= \frac{0.0015}{0.0015 + 0.84575} = \frac{0.0015}{0.84725} \approx 0.00177 \end{aligned}$$

- **Answer:** There is approximately a **0.18%** chance that an email in your main inbox is actually spam.

13 What is Statistical Independence?

13.1 The Intuitive Definition

Two events, A and B, are considered **independent** if the occurrence of one event provides no information about the likelihood of the other event occurring. Knowing that event A happened does not change your estimate of the probability of event B happening, and vice-versa.

13.2 The Formal Mathematical Definition

We can express this intuition formally using the language of conditional probability. Events A and B are independent if:

$$P(A|B) = P(A)$$

This reads: "The probability of A, given that B has occurred, is still just the probability of A." Symmetrically, it is also true that $P(B|A) = P(B)$.

14 The Key Consequence: Multiplication Rule for Independent Events

The most powerful and frequently used result of independence is a simplified rule for calculating the probability of both events occurring simultaneously.

The Multiplication Rule for Independent Events

If two events A and B are independent, then the probability of both A and B occurring (their intersection) is the product of their individual probabilities:

$$P(A \cap B) = P(A) \times P(B)$$

Derivation of the Rule

The derivation is straightforward and comes directly from the definition of conditional probability.

1. Recall the general multiplication rule: $P(A \cap B) = P(A|B) \times P(B)$.
2. By the definition of independence, we know that for independent events, $P(A|B) = P(A)$.
3. Substitute this into the rule: $P(A \cap B) = P(A) \times P(B)$.

15 Illustrative Example: A Deck of Cards

Independence with Playing Cards

Let's consider drawing a single card from a standard 52-card deck.

- **Event A:** The card drawn is a **Spade**.
- **Event B:** The card drawn is a **Queen**.

Question: Are these events independent?

1. **Calculate Individual Probabilities:**
   - $P(A) = P(\text{Spade}) = \frac{13}{52} = \frac{1}{4}$
   - $P(B) = P(\text{Queen}) = \frac{4}{52} = \frac{1}{13}$
2. **Use the Multiplication Rule (Assuming Independence):** The probability of drawing the Queen of Spades ($A \cap B$) should be:

$$P(A \cap B) = P(A) \times P(B) = \frac{1}{4} \times \frac{1}{13} = \frac{1}{52}$$

3. **Verify with the Sample Space:** There is exactly one "Queen of Spades" in the deck. Therefore, the direct probability is $P(\text{Queen of Spades}) = \frac{1}{52}$.

Since the results match, the events are indeed independent.

16 Application: System Reliability

The concept of independence is fundamental in engineering for analyzing the reliability of systems, assuming that the failure of one component is independent of others.

16.1 Components in Series

Imagine a system where if one component fails, the entire system fails.

- **Setup:** n components connected one after another.
- **Failure Condition:** The system fails if **any single component** fails.
- Let p be the probability that a single component fails.

The system succeeds only if **all** components succeed. The probability of a single component succeeding is $(1 - p)$.

$$P(\text{System Success}) = (1-p) \times (1-p) \times \cdots \times (1-p) = (1-p)^n$$
$$P(\text{Series Failure}) = 1 - P(\text{System Success}) = 1 - (1-p)^n$$

Example: For $n = 10$ components with $p = 0.05$ failure rate:

$$P(\text{Series Failure}) = 1 - (0.95)^{10} \approx 1 - 0.5987 \approx 0.4013$$

There is a **40.1% chance** the system will fail. Series systems decrease overall reliability.

16.2 Components in Parallel

Imagine a redundant system that works as long as at least one component is functioning.

- **Setup:** n components providing redundancy.
- **Failure Condition:** The system fails only if **all components fail**.
- Let p be the probability that a single component fails.

Since failures are independent, the probability of system failure is:

$$P(\text{Parallel Failure}) = p \times p \times \cdots \times p = p^n$$

Example: For $n = 10$ components with $p = 0.05$ failure rate:

$$P(\text{Parallel Failure}) = (0.05)^{10} \approx 9.77 \times 10^{-14}$$

This is an astronomically low probability. Parallel systems dramatically increase reliability.

Summary and Key Takeaways

- **Definition:** Two events are independent if knowing one occurred does not change the probability of the other ($P(A|B) = P(A)$).
- **Key Tool:** For independent events, $P(A \cap B) = P(A)P(B)$.
- **Application:** Crucial for analyzing IID (Independent and Identically Distributed) trials and is a cornerstone of reliability engineering.
- **Caveat:** The assumption of independence must be carefully scrutinized. Are the components of a system truly independent? A power surge could cause simultaneous failure, violating the assumption.

17 Practice Exercises

1. **Dice and Coins:** You roll a fair six-sided die and flip a fair coin. What is the probability that you roll an even number AND the coin lands on tails?
2. **Manufacturing Defects:** A factory produces widgets on two independent assembly lines. Line A has a 2% defect rate. Line B has a 3% defect rate. If you randomly select one widget from each line, what is the probability that both widgets are free of defects?

3. **Redundant Alarms:** A facility is protected by two independent fire alarms. System 1 has a 95% chance of detecting a fire. System 2 has a 98% chance. If a fire occurs, what is the probability that at least one alarm detects it?

18 Solutions to Exercises

18.1 Solution to Exercise 1 (Dice and Coins)

- Event A: Rolling an even number. $P(A) = 3/6 = 1/2$.
- Event B: Coin lands on tails. $P(B) = 1/2$.
- The events are independent.
- $P(A \cap B) = P(A) \times P(B) = (1/2) \times (1/2) = 1/4$.

18.2 Solution to Exercise 2 (Manufacturing Defects)

- Event A: Widget from Line A is NOT defective. $P(A) = 1 - 0.02 = 0.98$.
- Event B: Widget from Line B is NOT defective. $P(B) = 1 - 0.03 = 0.97$.
- The events are independent.
- $P(A \cap B) = P(A) \times P(B) = 0.98 \times 0.97 = 0.9506$.

18.3 Solution to Exercise 3 (Redundant Alarms)

It is easier to calculate the probability that *both alarms fail* and use the complement rule.

- Event A: System 1 fails. $P(A) = 1 - 0.95 = 0.05$.
- Event B: System 2 fails. $P(B) = 1 - 0.98 = 0.02$.
- The failures are independent, so $P(\text{Both Fail}) = P(A) \times P(B) = 0.05 \times 0.02 = 0.001$.
- The probability that *at least one* detects the fire is the complement of both failing:

  $$P(\text{At least one detects}) = 1 - P(\text{Both Fail}) = 1 - 0.001 = 0.999$$

  There is a 99.9% chance of detection.